QUADRATIC FORMS WITH FIVE LARGE EIGENVALUES HAVE BOUNDED DENSITIES

YITZCHAK SHMALO

ABSTRACT. Let $X_1, \dots, X_n$ be independent real random variables whose densities are bounded by K, and let $Q = X^T A X + b^T X + c$, where A is a real symmetric matrix. We show that if A has at least five eigenvalues of modulus at least $s > 0$, then Q has a density bounded by $270\,K^2/s$, and that if A has at least three such eigenvalues, then $\mathbb{E}|Q-a|^{-1/2} \le 77\,K/\sqrt{s}$ for every real a. The bounds do not depend on n, on the other eigenvalues, on b and c, or on the laws of the X_i, and no moment, smoothness or support assumptions are made. The analytic ingredient is a decay estimate for the characteristic function of a quadratic polynomial in terms of the ℓ-th elementary symmetric function of the squared singular values of an off-diagonal block. It is combined with a random sign argument showing that when all entries of A are small, $2m-1$ eigenvalues of modulus at least one force m singular values of some off-diagonal block to be bounded below, and with conditioning on one or two coordinates otherwise. At least three large eigenvalues are necessary for a density bound.

1. INTRODUCTION

Let $X = (X_1, \dots, X_n)$ be a random vector with independent coordinates, where X_i has a density f_i with $f_i \le K$ almost everywhere. For a real symmetric matrix A, a vector $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$ we consider the quadratic polynomial

$$Q = X^T A X + b^T X + c.$$

For linear forms the density question has a complete answer: if $|a| = 1$ then $a \cdot X$ has density at most $\sqrt{2}\,K$, by Rogozin's convolution inequality [10] combined with Ball's bound for sections of the cube [1]; see [11, 9] for projections of higher rank. For quadratic polynomials a bound on the eigenvalues of A from below does not by itself bound the density. When X_1, X_2 are uniform on $[0,1]$, the variable X_1^2 has density $(2\sqrt{y})^{-1}$ on $(0,1)$, and $X_1^2 - X_2^2 = (X_1 - X_2)(X_1 + X_2)$ has a density with a logarithmic singularity at 0. So one or two large eigenvalues are not enough. With three, two of them have the same sign, and for diagonal forms this already suffices: if $\lambda_1, \lambda_2 \ge 1$ then $\lambda_1 X_1^2 + \lambda_2 X_2^2$ has density at most $\int_0^1 u^{-1/2}(1-u)^{-1/2}\,\mathrm{d}u = \pi$ (each summand has density at most $y^{-1/2}$ on $(0,\infty)$), and adding independent terms cannot increase the supremum of a density. In general the eigenvectors of A are not aligned with the coordinates, and the coordinates of X in an eigenbasis are not independent; this is where the difficulty lies. It is natural to ask whether three eigenvalues of modulus at least one always give a bounded density, with a constant depending on nothing else. The main result of this paper shows that five do.

For a real matrix T we write $s_1(T) \ge s_2(T) \ge \cdots \ge 0$ for its singular values, padded with zeros. For symmetric A these are the moduli of the eigenvalues, so the hypothesis below says that A has at least five eigenvalues λ with $|\lambda| \ge s$.

Theorem 1.1. *Let $X_1, \dots, X_n$ be independent real random variables with densities bounded by K. Let $A \in \mathbb{R}^{n \times n}$ be symmetric, $b \in \mathbb{R}^n$ and $c \in \mathbb{R}$. If $s_5(A) \ge s > 0$, then $Q = X^T A X + b^T X + c$ has a density p_Q with*

$$\|p_Q\|_\infty \le \frac{270\,K^2}{s}. \tag{1.1}$$

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Equivalently, $\mathbb{P}(Q \in J) \le 270\,K^2|J|/s$ *for every interval* $J \subset \mathbb{R}$.

The dimension n, the remaining eigenvalues, the signature of A, the vector b and the scalar c are arbitrary, and the laws of the X_i need not be identical, smooth, compactly supported or have moments. Since $s_m(A) \ge s$ implies $s_5(A) \ge s$ for $m \ge 5$, the same bound holds under any larger multiplicity. The factor K^2/s is forced by scaling (Section 7), and the numerical constant has not been optimized.

The proof needs anti-concentration of a quadratic polynomial at a random point, and under three large eigenvalues we obtain it in the form of a negative half moment.

Theorem 1.2. *Under the assumptions of Theorem 1.1 with* $s_5(A) \ge s$ *replaced by* $s_3(A) \ge s > 0$*,*

$$\sup_{a\in\mathbb{R}} \mathbb{E}|Q-a|^{-1/2} \le \frac{77\,K}{\sqrt{s}}. \tag{1.2}$$

One large eigenvalue is not enough for (1.2): if X_1 is uniform on $[0,1]$, then $\mathbb{E}|X_1^2|^{-1/2} = \mathbb{E}X_1^{-1} = \infty$.

Outline. After scaling we may take $K = s = 1$. Fix $\eta = 1/12$ and $\zeta = 1/256$, and call a diagonal entry of modulus at least η, or a principal 2×2 minor of modulus at least ζ, a pivot.

If A has a pivot, we condition on all coordinates outside it. What remains is a quadratic polynomial in one or two variables whose density has a square-root or logarithmic singularity, located at a random point. That random point is again a quadratic polynomial, in the other coordinates, and its matrix is the Schur complement of the pivot, which still has at least four eigenvalues of modulus at least one after a diagonal pivot and at least three after a pair. We therefore need anti-concentration of a quadratic polynomial at the scale of the singularity: a negative half moment $\mathbb{E}|R-a|^{-1/2}$ for the square root, which is Theorem 1.2, and a logarithmic moment for the logarithm. Theorem 1.2 is proved by the same method one level down.

If A has no pivot, all its entries are small, and the spectral hypothesis has to be converted into a statement about an off-diagonal block. For a partition $[n] = I \sqcup J$ and $M = A_{I,J}$ let

$$E_\ell(M) = \sum_{\substack{S\subset I,\ T\subset J\\ |S|=|T|=\ell}} \det(M_{S,T})^2 = e_\ell(MM^T),$$

the ℓ-th elementary symmetric function of the squared singular values of M. The analytic core of the paper is Theorem 4.1: for every $\ell \ge 1$ and every real $t \ne 0$,

$$|\mathbb{E}e^{itQ}|^2 \le (\sqrt{2}\,\pi e)^\ell \Big(\frac{\ell^\ell}{\ell!}\Big)^{1/2} \frac{1}{E_\ell(A_{I,J})^{1/2}\,|t|^\ell},$$

uniformly in the other entries of A, in b, c, and in the laws of the X_i subject to $f_i \le 1$. For $\ell = 3$ the right side decays like $|t|^{-3/2}$ and Fourier inversion gives the density bound; $\ell = 2$ gives the negative half moment and $\ell = 1$ the logarithmic moment. The spectral hypothesis enters through Proposition 6.4: if A has no pivot and $s_{2m-1}(A) \ge 1$ with $m \le 3$, then some partition has $s_m(A_{I,J})$ bounded below by an absolute constant, and hence $E_m(A_{I,J})$ as well. Its proof flips the signs of a random set of coordinates. If D is the diagonal matrix of the signs, then $A - DAD$ is twice the off-diagonal block part of A for the corresponding partition. Compressing this identity to the span of $2m-1$ eigenvectors of A leaves the diagonal matrix of their eigenvalues, perturbed by an error whose mean square over the signs is small when the entries of A are small. The off-diagonal block part has the singular values of $A_{I,J}$, each twice, which turns $2m-1$ eigenvalues into m singular values. The count $5 = 2\cdot 3 - 1$ comes from this step at $m = 3$, and Remark 8.2 shows that four eigenvalues without pivots can leave every off-diagonal block of rank two.

Theorem 4.1 is proved by decoupling the two blocks with the Cauchy–Schwarz inequality, followed by a Hölder inequality over ℓ-sets of rows weighted by their Gram determinants and

a bound for densities of projections of product measures, which we derive from the geometric Brascamp–Lieb inequality. The Hölder step needs every row to carry at most half of the weight; a row carrying more is split off and integrated by Plancherel's theorem instead, and an entropy inequality (Lemma 4.2) controls the weights that remain.

Related work. Banks, Garza Vargas, Kulkarni and Srivastava [2, Theorem 2.1] bound the density of $\mathbf{X}^T Z \mathbf{Y} + \mathbf{X}^T U + V^T \mathbf{Y} + W$ for independent random matrices $\mathbf{X}, \mathbf{Y}$ whose entries are independent with bounded densities, in terms of the singular values of Z. There the two blocks enter bilinearly. For a general symmetric A the blocks $A_{I,I}$ and $A_{J,J}$ contribute quadratic terms within each block, and these are the terms that the argument of Section 4 has to tolerate. The geometric Brascamp–Lieb inequality is due to Ball [1] and Barthe [3]; the heat-flow proof we give follows Carlen, Lieb and Loss [6] and Bennett, Carbery, Christ and Tao [4]. Projection bounds for product densities appear in Rudelson and Vershynin [11], with sharp constants in Livshyts, Paouris and Pivovarov [9]. Small-ball estimates for polynomials go back to Carbery and Wright [5] for log-concave measures, and Costello, Tao and Vu [7] proved Littlewood–Offord type bounds for quadratic forms in discrete variables. Replacing a quadratic form by its off-diagonal part through independent random signs is the basic step of decoupling inequalities; see the monograph of de la Peña and Giné [8].

Organization. Section 2 collects Fourier-analytic and linear-algebraic preliminaries. Section 3 proves the geometric Brascamp–Lieb inequality and the projection and Fourier-power bounds derived from it. Section 4 proves Theorem 4.1. Section 5 contains the estimates for one and two coordinates, Section 6 the random sign argument, and Section 7 the proofs of Theorems 1.1 and 1.2. Section 8 discusses the threshold.

2. Preliminaries

2.1. **Fourier analysis.** For $f \in L^1(\mathbb{R}^d)$ we write $\widehat{f}(\xi) = \int_{\mathbb{R}^d} e^{i\xi\cdot x} f(x)\,\mathrm{d}x$, and for a real random variable Z we write $\phi_Z(t) = \mathbb{E}e^{itZ}$. If X_i has density f_i we put $\psi_i = \widehat{f_i} = \phi_{X_i}$. We write $\log_+ u = \max(0, \log u)$ for $u > 0$.

Lemma 2.1. (a) *If $f \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$, then $\int_{\mathbb{R}^d} |\widehat{f}|^2 = (2\pi)^d \int_{\mathbb{R}^d} |f|^2$.*

(b) *If $\phi_Z \in L^1(\mathbb{R})$, then Z has a continuous density $p_Z(x) = (2\pi)^{-1} \int e^{-itx}\phi_Z(t)\,\mathrm{d}t$, and $\|p_Z\|_\infty \le (2\pi)^{-1}\|\phi_Z\|_1$.*

(c) *For every real random variable Z, every $a \in \mathbb{R}$ and every $v > 0$,*

$$\mathbb{E}e^{-v(Z-a)^2} = \frac{1}{2\sqrt{\pi v}} \int_{\mathbb{R}} e^{-t^2/(4v)} e^{-ita}\phi_Z(t)\,\mathrm{d}t.$$

Proof. Parts (a) and (b) are Plancherel's theorem and the Fourier inversion theorem in the present normalization. For (c), the Gaussian integral $\int_{\mathbb{R}} e^{itx} e^{-t^2/(4v)}\,\mathrm{d}t = 2\sqrt{\pi v}\, e^{-vx^2}$ gives $e^{-v(z-a)^2} = (2\sqrt{\pi v})^{-1} \int e^{-t^2/(4v)} e^{it(z-a)}\,\mathrm{d}t$ for every real z; take $z = Z$, take expectations and apply Fubini's theorem, which is justified because the integrand is bounded by the integrable function $e^{-t^2/(4v)}$. □

Lemma 2.2. *If $a > 0$, $p > 1$ and $|\phi_Z(t)| \le \min\{1, (a/|t|)^p\}$ for all $t \ne 0$, then Z has a continuous density with $\|p_Z\|_\infty \le ap/(\pi(p-1))$.*

Proof. The envelope has integral $2a + 2a^p \int_a^\infty t^{-p}\,\mathrm{d}t = 2ap/(p-1)$; apply Lemma 2.1(b). □

Lemma 2.3. *For every real random variable Z and every $a \in \mathbb{R}$,*

$$\mathbb{E}|Z-a|^{-1/2} \le \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} |t|^{-1/2} |\phi_Z(t)|\,\mathrm{d}t, \tag{2.1}$$

where both sides may be infinite and $|0|^{-1/2} = \infty$. In particular, if $|\phi_Z(t)| \le \min\{1, u/|t|\}$ for all $t \ne 0$, then $\sup_a \mathbb{E}|Z-a|^{-1/2} \le 8\sqrt{u/(2\pi)}$.

Proof. For real z we have $|z|^{-1/2} = \Gamma(1/4)^{-1} \int_0^\infty v^{-3/4} e^{-vz^2}\,\mathrm{d}v$, including $z = 0$ with value $+\infty$. By Tonelli's theorem and Lemma 2.1(c),

$$\mathbb{E}|Z-a|^{-1/2} \le \frac{1}{\Gamma(1/4)} \int_0^\infty \frac{v^{-3/4}}{2\sqrt{\pi v}} \int_{\mathbb{R}} e^{-t^2/(4v)} |\phi_Z(t)|\,\mathrm{d}t\,\mathrm{d}v.$$

Exchanging the integrals (Tonelli again) and substituting $w = t^2/(4v)$ gives

$$\int_0^\infty v^{-5/4} e^{-t^2/(4v)}\,\mathrm{d}v = 4^{1/4}\,\Gamma(1/4)\,|t|^{-1/2} \qquad (t \neq 0),$$

and (2.1) follows since $4^{1/4}/(2\sqrt{\pi}) = 1/\sqrt{2\pi}$. Under the stated envelope,

$$\int_{\mathbb{R}} |t|^{-1/2} \min\{1, u/|t|\}\,\mathrm{d}t = 4\sqrt{u} + 4\sqrt{u} = 8\sqrt{u}.$$

□

2.2. **Conditioning.** All conditioning in this paper is on original coordinates, and it enters through the following domination principle.

Lemma 2.4. *Let U and V be independent random elements of measurable spaces $\mathcal{U}$ and $\mathcal{V}$, let $h : \mathcal{U} \times \mathcal{V} \to \mathbb{R}$ be measurable, and let $g : \mathbb{R} \times \mathcal{V} \to [0,\infty]$ be measurable. Suppose that for almost every v with respect to the law of V, $\mathbb{P}(h(U,v) \in B) \le \int_B g(z,v)\,\mathrm{d}z$ for every Borel set B. Then $Q = h(U,V)$ has a density p_Q with $p_Q(z) \le \mathbb{E}g(z,V)$ for almost every z, and $\mathbb{E}k(Q) \le \int_{\mathbb{R}} k(z)\,\mathbb{E}g(z,V)\,\mathrm{d}z$ for every Borel $k \ge 0$.*

Proof. By independence and Tonelli's theorem, $\mathbb{P}(Q \in B) = \int \mathbb{P}(h(U,v) \in B)\,\mathrm{d}\mathbb{P}_V(v) \le \int_B \mathbb{E}g(z,V)\,\mathrm{d}z$. The right side vanishes when B is Lebesgue null, so Q has a density, and the inequality for all B gives $p_Q \le \mathbb{E}g(\cdot,V)$ almost everywhere. The last assertion follows by integrating k against p_Q. □

2.3. **Linear algebra.** For a matrix T, $\|T\|$ is the operator norm and $\|T\|_F$ the Frobenius norm. For symmetric A and $t > 0$ let $N_A(t) = \#\{j : |\lambda_j(A)| \ge t\}$, so that $N_A(t) \ge m$ if and only if $s_m(A) \ge t$. For index sets S, T we write $A_{S,T}$ for the submatrix with rows in S and columns in T, in increasing order, and M_T for the rows of M indexed by T.

Lemma 2.5. *For real matrices of the same size, $|s_j(T) - s_j(U)| \le \|T-U\|$, and $s_{j+q}(T+R) \le s_j(T)$ whenever $\operatorname{rank} R \le q$. If U and V have orthonormal columns and U^TTV is defined, then $s_j(U^TTV) \le s_j(T)$. No symmetry is required.*

Proof. Both follow from $s_j(T) = \min\{\|T - B\| : \operatorname{rank} B < j\}$. The minimum is attained by truncating the singular value decomposition after $j-1$ terms; conversely, if $\operatorname{rank} B < j$, the kernel of B meets the span of the first j right singular vectors of T in a unit vector x, and $\|(T-B)x\| = \|Tx\| \ge s_j(T)$. If $\operatorname{rank} B < j$ then $\|T - B\| \le \|U - B\| + \|T - U\|$, which gives the first assertion. If $\operatorname{rank} B < j$ and $\operatorname{rank} R \le q$ then $\operatorname{rank}(B+R) < j+q$ and $\|T + R - (B+R)\| = \|T - B\|$, which gives the second. If $\operatorname{rank} B < j$ then $\operatorname{rank}(U^TBV) < j$ and $\|U^TTV - U^TBV\| \le \|T - B\|$, which gives the third. □

Lemma 2.6. *Let $A = \begin{pmatrix} B & C \\ C^T & D \end{pmatrix}$ be real symmetric with $B \in \mathbb{R}^{k\times k}$ invertible, and let $S = D - C^TB^{-1}C$. Then $N_S(t) \ge N_A(t) - k$ for every $t > 0$.*

Proof. Let n be the size of A. Suppose first that A is invertible. Then $\det A = \det B \det S$, so S is invertible, and the block inversion formula shows that the lower right block of A^{-1} is S^{-1}. By Cauchy's interlacing theorem, the j-th largest eigenvalue of this principal submatrix is at most the j-th largest eigenvalue of A^{-1}; hence S^{-1} has at most as many eigenvalues larger than $1/t$ as A^{-1}, and, applying the same to $-A^{-1}$, at most as many smaller than $-1/t$. Since the eigenvalues of S^{-1} and A^{-1} are the reciprocals of those of S and A,

$$(n-k) - N_S(t) = \#\{j : |\lambda_j(S)| < t\} \le \#\{j : |\lambda_j(A)| < t\} = n - N_A(t).$$

In general, for all but finitely many $\varepsilon \in (-t,t)$ both $A+\varepsilon I$ and $B+\varepsilon I$ are invertible. The Schur complement $S_\varepsilon = D+\varepsilon I - C^T(B+\varepsilon I)^{-1}C$ tends to S as $\varepsilon \to 0$, and $N_{A+\varepsilon I}(t-|\varepsilon|) \geq N_A(t)$. With $m = N_A(t) - k \geq 1$ (otherwise there is nothing to prove), the invertible case gives $s_m(S_\varepsilon) \geq t - |\varepsilon|$, and Lemma 2.5 gives $s_m(S) \geq t$ in the limit. □

Lemma 2.7. *Let $M \in \mathbb{R}^{I\times J}$ have rows $m_i \in \mathbb{R}^J$, and write $\Theta(T) = \det(M_T M_T^T)$ for $T \subset I$, with $\Theta(\varnothing) = 1$.*

(a) *For $\ell \geq 1$, $e_\ell(MM^T) = \sum_{T\subset I,\ |T|=\ell} \Theta(T) = \sum_{S\subset I,\ T\subset J,\ |S|=|T|=\ell} \det(M_{S,T})^2$, where e_ℓ denotes the ℓ-th elementary symmetric function of the eigenvalues.*

(b) *Let $H \subset I$ with $\Theta(H) > 0$, let Π be the orthogonal projection of $\mathbb{R}^J$ onto the span of $\{m_h : h \in H\}$, and let $c_i = (I-\Pi)m_i$. For $T \subset I \setminus H$ let C_T have rows c_i, $i \in T$. Then $\Theta(H \cup T) = \Theta(H)\det(C_T C_T^T)$.*

Proof. (a) The coefficient of z^ℓ in $\det(I + zMM^T)$ is both $e_\ell(MM^T)$ and the sum of the principal $\ell\times\ell$ minors of MM^T, which are the $\Theta(T)$; by the Cauchy–Binet formula $\Theta(T) = \sum_{S'\subset J,\ |S'|=\ell} \det(M_{T,S'})^2$. (b) The Gram matrix of the rows in $H\cup T$ is $\begin{pmatrix} G_{HH} & G_{HT} \\ G_{TH} & G_{TT}\end{pmatrix}$ with $G_{HH} = M_H M_H^T$ invertible. Its determinant is $\det G_{HH}\det(G_{TT} - G_{TH}G_{HH}^{-1}G_{HT})$. Since $\Pi = M_H^T G_{HH}^{-1} M_H$, the Schur complement equals $M_T(I-\Pi)M_T^T = C_T C_T^T$. □

3. Projections of product densities

Theorem 3.1 (geometric Brascamp–Lieb inequality)**.** *Let $u_1,\dots,u_N$ be unit vectors in $\mathbb{R}^d$ and $c_1,\dots,c_N > 0$ with $\sum_i c_i u_i u_i^T = I_d$. For nonnegative integrable functions $h_1,\dots,h_N$ on $\mathbb{R}$,*

$$\int_{\mathbb{R}^d} \prod_{i=1}^N h_i(u_i\cdot x)^{c_i}\,\mathrm{d}x \leq \prod_{i=1}^N \Big(\int_{\mathbb{R}} h_i\Big)^{c_i}.$$

Proof. Taking traces in the frame identity gives $\sum_i c_i = d$. Suppose first that each h_i is bounded, vanishes outside $[-m,m]$ and has $\mu_i = \int h_i > 0$. For $\tau > 0$ let $g_\tau(y) = (4\pi\tau)^{-1/2}e^{-y^2/(4\tau)}$, let $f_i(\tau,\cdot) = g_\tau * h_i$, and put

$$F(\tau,x) = \prod_i f_i(\tau, u_i\cdot x)^{c_i}.$$

Each f_i is smooth and positive, and $\partial_\tau f_i = \partial_y^2 f_i$. Writing $v_i = \partial_y \log f_i(\tau, u_i\cdot x)$ and using $|u_i| = 1$, a direct computation gives

$$\frac{\partial_\tau F - \Delta_x F}{F} = \sum_i c_i v_i^2 - \Big|\sum_i c_i v_i u_i\Big|^2 \geq 0.$$

The inequality holds because the linear map $W : \mathbb{R}^N \to \mathbb{R}^d$ with columns $\sqrt{c_i}\,u_i$ satisfies $WW^T = I_d$, so $\|W\| = 1$, and $\sum_i c_i v_i u_i = W(\sqrt{c_i}\,v_i)_i$.

We justify $\frac{\mathrm{d}}{\mathrm{d}\tau}\int F\,\mathrm{d}x = \int \partial_\tau F\,\mathrm{d}x \geq \int \Delta_x F\,\mathrm{d}x = 0$ on a compact interval $\tau \in [\tau_0,\tau_1] \subset (0,\infty)$. There $f_i(\tau,y) \leq 2m\|h_i\|_\infty \sup_{|z|\leq m} g_\tau(y-z) \leq Ce^{-y^2/(16\tau_1)}$ (for $|y| \geq 2m$ use $|y-z| \geq |y|/2$), and $\partial_y \log f_i$ is the average of $-(y-z)/(2\tau)$ over $z \in [-m,m]$ against a probability measure, so $|\partial_y \log f_i(\tau,y)| \leq (|y|+m)/(2\tau_0)$; similar bounds hold for the second derivative. Using $\sum_i c_i(u_i\cdot x)^2 = |x|^2$, the functions F, $\partial_\tau F$ and the first two x-derivatives of F are bounded by $C(1+|x|^4)e^{-|x|^2/(16\tau_1)}$ uniformly in $\tau \in [\tau_0,\tau_1]$. This justifies differentiating under the integral sign and shows $\int \Delta_x F\,\mathrm{d}x = 0$. Thus $\int F(\tau,x)\,\mathrm{d}x$ is nondecreasing in τ.

As $\tau\to\infty$, for fixed $z \in \mathbb{R}^d$ we have $\sqrt{\tau}\,f_i(\tau,\sqrt{\tau}\,u_i\cdot z) \to (4\pi)^{-1/2}\mu_i e^{-(u_i\cdot z)^2/4}$. Substituting $x = \sqrt{\tau}z$ and using $\sum c_i = d$,

$$\int F(\tau,x)\,\mathrm{d}x = \int \prod_i \big(\sqrt{\tau}f_i(\tau,\sqrt{\tau}u_i\cdot z)\big)^{c_i}\,\mathrm{d}z \longrightarrow (4\pi)^{-d/2}\prod_i \mu_i^{c_i}\int e^{-|z|^2/4}\,\mathrm{d}z = \prod_i \mu_i^{c_i},$$

by dominated convergence, since for $\tau \geq 1$ the integrand is at most $Ce^{-|z|^2/16}$.

As $\tau \downarrow 0$, $f_i(\tau, y) \to h_i(y)$ at every Lebesgue point y of h_i. The preimage of a Lebesgue null set of $\mathbb{R}$ under $x \mapsto u_i \cdot x$ is null in $\mathbb{R}^d$, so for almost every x each $u_i \cdot x$ is a Lebesgue point of h_i, and $F(\tau, x) \to \prod_i h_i(u_i \cdot x)^{c_i}$. Fatou's lemma and the monotonicity give the inequality in this case.

If some h_i vanishes almost everywhere, the left side is zero for the same reason. For general h_i apply the case just proved to $\min(h_i, m)\mathbf{1}_{[-m,m]}$ and let $m \to \infty$, using monotone convergence on both sides. □

Lemma 3.2. *Let $X \in \mathbb{R}^n$ have independent coordinates with densities bounded by 1.*

(a) *If $U \in \mathbb{R}^{n\times d}$ satisfies $U^T U = I_d$, then $U^T X$ has a density bounded by $e^{d/2}$.*

(b) *If $B \in \mathbb{R}^{d\times n}$ has rank d, then BX has a density bounded by $e^{d/2}\det(BB^T)^{-1/2}$.*

In particular $a \cdot X$ has density at most $\sqrt{e}/|a|$ for every $a \neq 0$.

Proof. (a) Choose versions of the densities f_i with $0 \le f_i \le 1$ everywhere. If $d = n$, the density of $U^T X$ is $\prod_i f_i((Uy)_i) \le 1$. If $d < n$, complete U to an orthogonal matrix (U, W) and let w_i^T be the i-th row of W. Since $(y, z) \mapsto Uy + Wz$ is orthogonal, $U^T X$ has density

$$p(y) = \int_{\mathbb{R}^{n-d}} \prod_{i=1}^{n} f_i\big((Uy)_i + w_i \cdot z\big)\, \mathrm{d}z.$$

Let $\beta_i = |w_i|^2$. Then $\sum_i w_i w_i^T = W^T W = I_{n-d}$ and $\sum_i \beta_i = n - d$. Factors with $\beta_i = 0$ are at most one and are dropped. For the others put $v_i = w_i/\sqrt{\beta_i}$, so that $\sum \beta_i v_i v_i^T = I_{n-d}$, and $h_i(\tau) = f_i((Uy)_i + \sqrt{\beta_i}\tau)^{1/\beta_i}$. Theorem 3.1 with exponents β_i gives $p(y) \le \prod_i (\int h_i)^{\beta_i}$. Since $0 < \beta_i \le 1$ and $f_i \le 1$ we have $f_i^{1/\beta_i} \le f_i$, so $\int h_i \le \beta_i^{-1/2}$ and

$$\log p(y) \le \frac{1}{2}\sum_{i=1}^{n} \beta_i \log\frac{1}{\beta_i} \le \frac{1}{2}\sum_{i=1}^{n}(1 - \beta_i) = \frac{d}{2},$$

where the terms with $\beta_i = 0$ are read as 0 in the first sum, and we used $\beta\log(1/\beta) \le 1 - \beta$ for $0 < \beta \le 1$, which follows from $\log x \le x - 1$ with $x = 1/\beta$ (for $\beta_i = 0$ the second inequality is $0 \le 1$).

(b) Write $B = P\Sigma U^T$ with $P \in \mathbb{R}^{d\times d}$ orthogonal, Σ diagonal with positive entries and $U^T U = I_d$. Then $BX = P\Sigma(U^T X)$, and a linear change of variables with Jacobian $\det\Sigma = \det(BB^T)^{1/2}$ transfers the bound of (a). The last assertion is (b) with $d = 1$. □

Lemma 3.3. *Let f be a probability density with $f \le 1$. Then for every real $p \ge 2$,*

$$\int_{\mathbb{R}} |\widehat{f}(t)|^p \,\mathrm{d}t \le \frac{2\pi\sqrt{2e}}{\sqrt{p}}.$$

If Y, Y' are independent with density f, then $Y - Y'$ has a density bounded by 1.

Proof. The density of $Y - Y'$ is $\int f(x + y)f(y)\,\mathrm{d}y \le \|f\|_\infty \|f\|_1 \le 1$. Let $m = \lfloor p/2 \rfloor$; then $m \ge 1$ and $m \ge p/4$. Since $|\widehat{f}| \le 1$, $\int|\widehat{f}|^p \le \int|\widehat{f}|^{2m}$. The function $|\widehat{f}|^{2m}$ is the characteristic function of $Z = \sum_{j=1}^{m}(Y_j - Y_j')$ for independent copies, and it is integrable because $\int|\widehat{f}|^{2m} \le \int|\widehat{f}|^2 = 2\pi\|f\|_2^2 \le 2\pi$. By Lemma 2.1(b), Z has a continuous density whose value at 0 is $(2\pi)^{-1}\int|\widehat{f}|^{2m}$. On the other hand $Z = \mathbf{1}\cdot(Y_1, \dots, Y_m, -Y_1', \dots, -Y_m')$ is a linear form with coefficient vector of length $\sqrt{2m}$ in $2m$ independent variables with densities bounded by one, so by Lemma 3.2 its density is at most $\sqrt{e}/\sqrt{2m}$ almost everywhere, hence everywhere for the continuous version. Therefore $\int|\widehat{f}|^p \le 2\pi\sqrt{e}/\sqrt{2m} \le 2\pi\sqrt{2e}/\sqrt{p}$. □

4. Fourier decay from crossing volumes

Throughout this section $X = (X_1, \dots, X_n)$ has independent coordinates with densities $f_i \le 1$, A is real symmetric, $b \in \mathbb{R}^n$, $c \in \mathbb{R}$, and $Q = X^T AX + b^T X + c$. We fix a partition

$[n] = I \sqcup J$ and put $M = A_{I,J} \in \mathbb{R}^{I\times J}$, with rows $m_i \in \mathbb{R}^J$ for $i \in I$. As in Lemma 2.7, $\Theta(T) = \det(M_T M_T^T)$ for $T \subset I$, and for $\ell \geq 1$ we define the crossing volume

$$E_\ell(M) = \sum_{T\subset I,\ |T|=\ell} \Theta(T) = e_\ell(MM^T) = \sum_{\substack{S\subset I,\ T\subset J\\ |S|=|T|=\ell}} \det(M_{S,T})^2.$$

For $H \subset I$ with $|H| \leq \ell$ we also put $\Theta_\ell(H) = \sum\{\Theta(T) : H \subset T \subset I,\ |T| = \ell\}$, so that $\Theta_\ell(\varnothing) = E_\ell(M)$. Finally $\omega(m) = m^m/m!$ for integers $m \geq 1$ and $\omega(0) = 1$.

Theorem 4.1. *If $\ell \geq 1$ and $E_\ell(M) > 0$, then for every real $t \neq 0$*

$$|\phi_Q(t)|^2 \leq \left(\sqrt{2}\,\pi e\right)^\ell \omega(\ell)^{1/2}\, E_\ell(M)^{-1/2}\, |t|^{-\ell}. \tag{4.1}$$

The bound is uniform in the blocks $A_{I,I}$ and $A_{J,J}$, in b and c, and in the laws of the X_i. It rests on two lemmas.

Lemma 4.2. *Let $m \geq 1$, let w be a probability distribution on the m-element subsets of a finite set, let $\alpha_i = \sum_{T\ni i} w_T$, and let $\mathrm{H}(w) = -\sum_T w_T \log w_T$. Then, with $0\log 0 = 0$,*

$$\mathrm{H}(w) + \sum_i \alpha_i \log \alpha_i \leq \log \omega(m).$$

Proof. If $w_T > 0$ then $\alpha_i \geq w_T > 0$ for every $i \in T$, and $\sum_i \alpha_i \log \alpha_i = \sum_T w_T \sum_{i\in T} \log \alpha_i$, the sums running over sets with $w_T > 0$. Hence the left side equals $\sum_{w_T>0} w_T \log\bigl(\prod_{i\in T} \alpha_i / w_T\bigr)$, which by concavity of the logarithm is at most $\log \sum_T \prod_{i\in T} \alpha_i$. In the expansion of $(\sum_i \alpha_i)^m$ each product $\prod_{i\in T} \alpha_i$ with $|T| = m$ occurs $m!$ times and all other terms are nonnegative, so $\sum_T \prod_{i\in T} \alpha_i \leq (\sum_i \alpha_i)^m/m! = m^m/m!$, because $\sum_i \alpha_i = m$. □

Lemma 4.3. *Let $\ell \geq 1$ and let $H \subset I$ with $k = |H| \leq \ell$ and $\Theta_\ell(H) > 0$. If $k < \ell$, assume in addition that*

$$\Theta_\ell(H \cup \{j\}) \leq \tfrac{1}{2}\Theta_\ell(H) \qquad \text{for every } j \in I \setminus H. \tag{4.2}$$

Then for every real $t \neq 0$

$$|\phi_Q(t)|^2 \leq \pi^\ell e^{\ell/2} (2e)^{(\ell-k)/2} \omega(\ell-k)^{1/2}\, \Theta_\ell(H)^{-1/2} |t|^{-\ell}. \tag{4.3}$$

Proof. Write $I' = I \setminus H$ and $\ell' = \ell - k$. Since $\Theta_\ell(H) > 0$, some $T \supset H$ has $\Theta(T) > 0$; hence the rows m_h, $h \in H$, are linearly independent and $\Theta(H) > 0$. Let Π be the orthogonal projection of $\mathbb{R}^J$ onto their span ($\Pi = 0$ if $k = 0$), and for $i \in I'$ write $m_i = c_i + M_H^T \lambda_i$ with $c_i = (I - \Pi)m_i$ and $\lambda_i \in \mathbb{R}^H$. For $T' \subset I'$ let $C_{T'}$ have rows c_i, $i \in T'$, and $\theta(T') = \det(C_{T'} C_{T'}^T)$. By Lemma 2.7(b), $\Theta(H \cup T') = \Theta(H)\theta(T')$, so

$$Z_H := \sum_{T'\subset I',\ |T'|=\ell'} \theta(T') = \frac{\Theta_\ell(H)}{\Theta(H)} > 0.$$

When $\ell' \geq 1$ we let $w_{T'} = \theta(T')/Z_H$, a probability distribution on the ℓ'-subsets of I', whose marginals satisfy

$$\alpha_i = \sum_{T'\ni i} w_{T'} = \frac{\Theta_\ell(H \cup \{i\})}{\Theta_\ell(H)} \leq \frac{1}{2} \qquad (i \in I')$$

by (4.2). Let X'_J be an independent copy of X_J, independent of X, and $D = X_J - X'_J$; by Lemma 3.3, D has independent coordinates with densities bounded by one. Split

$$Q = F(X_H) + 2X_H^T(A_{H,I'}X_{I'} + M_H X_J) + R(X_{I'}, X_J),$$

where $F(x_H) = x_H^T A_{H,H} x_H + b_H^T x_H$ and

$$R(x', y) = x'^T A_{I',I'} x' + 2x'^T A_{I',J}\, y + G(y) + b_{I'}^T x' + c, \qquad G(y) = y^T A_{J,J} y + b_J^T y.$$

The rows of $A_{I',J}$ are the m_i, $i \in I'$, and $R(x', y) - R(x', y') = 2x'^T A_{I',J}(y - y') + G(y) - G(y')$.

Step 1: the case $k = 0$. Here $H = \varnothing$, $I' = I$ and $Q = F_0(X_I) + 2X_I^T M X_J + G(X_J)$ with $F_0(x) = x^T A_{I,I} x + b_I^T x + c$. Integrating first over X_J and applying the Cauchy–Schwarz inequality in X_I,

$$|\phi_Q(t)|^2 \le \mathbb{E}_{X_I}\Big|\mathbb{E}_{X_J} e^{it(2X_I^T M X_J + G(X_J))}\Big|^2 = \mathbb{E}\Big[e^{it(G(X_J)-G(X_J'))}\prod_{i\in I}\psi_i\big(2t\langle m_i, D\rangle\big)\Big],$$

where we expanded the square with the copy X_J' and then integrated over the independent coordinates X_i, $i \in I$. All integrands are bounded, so Fubini's theorem applies. Hence

$$|\phi_Q(t)|^2 \le \mathbb{E}\prod_{i\in I}|\psi_i(2t\langle m_i, D\rangle)|. \tag{4.4}$$

Step 2: the case $k \ge 1$. Let $p_H(x_H) = \prod_{h\in H} f_h(x_h) \le 1$ and $a(x_H) = e^{itF(x_H)} p_H(x_H)$; then $\|a\|_2^2 \le \|p_H\|_1 = 1$. By independence,

$$\mathbb{E}[e^{itQ} \mid X_{I'}, X_J] = e^{itR(X_{I'},X_J)}\,\widehat{a}\big(2t(A_{H,I'}X_{I'} + Y)\big), \qquad Y = M_H X_J \in \mathbb{R}^H.$$

By Lemma 3.2(b), Y has a density $p_Y \le e^{k/2}\Theta(H)^{-1/2}$. For $x \in \mathbb{R}^{I'}$ let ν_x be the complex measure $\nu_x(B) = \mathbb{E}[e^{itR(x,X_J)}\mathbf{1}_B(Y)]$. Since $|\nu_x(B)| \le \mathbb{P}(Y \in B)$, it has a density, still written ν_x, with $|\nu_x| \le p_Y$; in particular $\|\nu_x\|_2^2 \le \|p_Y\|_\infty$. Conditionally on $X_{I'} = x$, the Cauchy–Schwarz inequality and Lemma 2.1(a) in $\mathbb{R}^H$ give

$$\Big|\int \widehat{a}(2t(A_{H,I'}x + y))\nu_x(y)\,\mathrm{d}y\Big|^2 \le \frac{(2\pi)^k\|a\|_2^2}{(2|t|)^k}\|\nu_x\|_2^2 \le \Big(\frac{\pi}{|t|}\Big)^k\|\nu_x\|_2^2,$$

and averaging over $X_{I'}$,

$$|\phi_Q(t)|^2 \le \Big(\frac{\pi}{|t|}\Big)^k \mathbb{E}\|\nu_{X_{I'}}\|_2^2. \tag{4.5}$$

If $\ell' = 0$ then $\Theta_\ell(H) = \Theta(H)$, and (4.5) with $\|\nu_x\|_2^2 \le e^{\ell/2}\Theta(H)^{-1/2}$ is (4.3). Assume $\ell' \ge 1$. Fix a continuous probability density ρ on $\mathbb{R}^H$ with compact support, let $\rho_\varepsilon(y) = \varepsilon^{-k}\rho(y/\varepsilon)$ and $\kappa_\varepsilon(z) = \int \rho_\varepsilon(y)\rho_\varepsilon(y+z)\,\mathrm{d}y$, a probability density. Since $(\nu_x * \rho_\varepsilon)(y) = \mathbb{E}[e^{itR(x,X_J)}\rho_\varepsilon(y - Y)]$, Fubini's theorem gives

$$\|\nu_x * \rho_\varepsilon\|_2^2 = \mathbb{E}\Big[e^{it(R(x,X_J)-R(x,X_J'))}\kappa_\varepsilon(M_H D)\Big],$$

which is a measurable function of x. As $\varepsilon \to 0$, $\|\nu_x * \rho_\varepsilon\|_2 \to \|\nu_x\|_2$ for every x because $\nu_x \in L^2$, and $\|\nu_x * \rho_\varepsilon\|_2^2 \le \|p_Y\|_\infty$; so by dominated convergence $\mathbb{E}\|\nu_{X_{I'}}\|_2^2 = \lim_{\varepsilon\to 0}\mathbb{E}\|\nu_{X_{I'}} * \rho_\varepsilon\|_2^2$. Replacing x by $X_{I'}$ and integrating over $X_{I'}$ first,

$$\mathbb{E}\|\nu_{X_{I'}} * \rho_\varepsilon\|_2^2 = \mathbb{E}\Big[e^{it(G(X_J)-G(X_J'))}\kappa_\varepsilon(M_H D)\prod_{i\in I'}\psi_i\big(2t\langle m_i, D\rangle\big)\Big].$$

Therefore

$$|\phi_Q(t)|^2 \le \Big(\frac{\pi}{|t|}\Big)^k \sup_{\varepsilon>0}\mathbb{E}\Big[\kappa_\varepsilon(M_H D)\prod_{i\in I'}|\psi_i(2t\langle m_i, D\rangle)|\Big]. \tag{4.6}$$

Inequality (4.4) is (4.6) with $k = 0$ and $\kappa_\varepsilon \equiv 1$, and we treat both cases together from now on.

Step 3: Hölder. Let $I'_+ = \{i \in I' : \alpha_i > 0\}$. If $w_{T'} > 0$ then $\alpha_i \ge w_{T'} > 0$ for all $i \in T'$. Since $|\psi_i| \le 1$, and since $\sum_{T'\ni i} w_{T'}/\alpha_i = 1$ for $i \in I'_+$,

$$\prod_{i\in I'}|\psi_i| \le \prod_{i\in I'_+}|\psi_i| = \prod_{T'}\Big(\prod_{i\in T'}|\psi_i|^{1/\alpha_i}\Big)^{w_{T'}},$$

the products over T' running over sets with $w_{T'} > 0$. As $\kappa_\varepsilon = \prod_{T'}\kappa_\varepsilon^{w_{T'}}$, Hölder's inequality with exponents $1/w_{T'}$ gives

$$\mathbb{E}\Big[\kappa_\varepsilon(M_H D)\prod_{i\in I'}|\psi_i(2t\langle m_i, D\rangle)|\Big] \le \prod_{T'}\Big(\mathbb{E}\Big[\kappa_\varepsilon(M_H D)\prod_{i\in T'}|\psi_i(2t\langle m_i, D\rangle)|^{1/\alpha_i}\Big]\Big)^{w_{T'}}.$$

Step 4: one set T'. Fix T' with $w_{T'} > 0$ and let B be the matrix with rows m_h ($h \in H$) and c_i ($i \in T'$). Its two groups of rows are orthogonal, so $\det(BB^T) = \Theta(H)\theta(T') > 0$, and by Lemma 3.2(b) the vector $(M_H D, (\langle c_i, D\rangle)_{i\in T'}) \in \mathbb{R}^H \times \mathbb{R}^{T'}$ has a density bounded by $e^{\ell/2}(\Theta(H)\theta(T'))^{-1/2}$. Since $\langle m_i, D\rangle = \langle c_i, D\rangle + \lambda_i \cdot M_H D$,

$$\mathbb{E}\Big[\kappa_\varepsilon(M_H D)\prod_{i\in T'}|\psi_i(2t\langle m_i, D\rangle)|^{1/\alpha_i}\Big] \le \frac{e^{\ell/2}}{(\Theta(H)\theta(T'))^{1/2}}\int_{\mathbb{R}^H}\kappa_\varepsilon(y)\prod_{i\in T'}\int_{\mathbb{R}}|\psi_i(2t(z+\lambda_i\cdot y))|^{1/\alpha_i}\,\mathrm{d}z\,\mathrm{d}y.$$

Each inner integral equals $(2|t|)^{-1}\int|\psi_i|^{1/\alpha_i}$, which by Lemma 3.3 with $p = 1/\alpha_i \ge 2$ is at most $\pi\sqrt{2e\alpha_i}/|t|$. Since $\int\kappa_\varepsilon = 1$, the right side is at most

$$e^{\ell/2}\big(\Theta(H)\theta(T')\big)^{-1/2}\Big(\frac{\pi\sqrt{2e}}{|t|}\Big)^{\ell'}\prod_{i\in T'}\alpha_i^{1/2}.$$

(When $k = 0$ the same computation applies without the variable y, with $\Theta(\varnothing) = 1$.)

Step 5: entropy. Taking the product over T' with weights $w_{T'}$ and writing $\theta(T') = Z_H w_{T'}$,

$$\prod_{T'}\big(\Theta(H)\theta(T')\big)^{-w_{T'}/2} = \big(\Theta(H)Z_H\big)^{-1/2}e^{\mathrm{H}(w)/2} = \Theta_\ell(H)^{-1/2}e^{\mathrm{H}(w)/2}$$

and

$$\prod_{T'}\prod_{i\in T'}\alpha_i^{w_{T'}/2} = e^{\frac12\sum_i\alpha_i\log\alpha_i}.$$

By Lemma 4.2, $e^{(\mathrm{H}(w)+\sum_i\alpha_i\log\alpha_i)/2} \le \omega(\ell')^{1/2}$. Combining Steps 3 and 4 with (4.6),

$$|\phi_Q(t)|^2 \le \Big(\frac{\pi}{|t|}\Big)^k e^{\ell/2}\Big(\frac{\pi\sqrt{2e}}{|t|}\Big)^{\ell'}\omega(\ell')^{1/2}\,\Theta_\ell(H)^{-1/2},$$

which is (4.3). □

Proof of Theorem 4.1. Put $H_0 = \varnothing$. Given H_r with $|H_r| = r < \ell$, if some $j \in I \setminus H_r$ has $\Theta_\ell(H_r \cup \{j\}) > \frac12\Theta_\ell(H_r)$, let $H_{r+1} = H_r \cup \{j\}$; otherwise stop. The procedure stops at some $H = H_k$ with $k \le \ell$, and by construction $\Theta_\ell(H) \ge 2^{-k}E_\ell(M) > 0$ and (4.2) holds if $k < \ell$. Lemma 4.3 gives

$$|\phi_Q(t)|^2 \le \pi^\ell e^{\ell/2}\,a_k\,E_\ell(M)^{-1/2}|t|^{-\ell}, \qquad a_k = (2e)^{(\ell-k)/2}\omega(\ell-k)^{1/2}2^{k/2}.$$

For $1 \le m \le \ell$ we have $\omega(m)/\omega(m-1) = (m/(m-1))^{m-1} \ge 1$ (read as 1 for $m = 1$), hence $a_{k+1}/a_k = e^{-1/2}(\omega(\ell-k-1)/\omega(\ell-k))^{1/2} \le e^{-1/2}$. Thus $a_k \le a_0 = (2e)^{\ell/2}\omega(\ell)^{1/2}$, and $\pi^\ell e^{\ell/2}(2e)^{\ell/2} = (\sqrt{2}\,\pi e)^\ell$. □

Corollary 4.4. *Let $E_\ell = E_\ell(M)$.*

(a) *If $E_3 > 0$, then Q has a continuous density with $\|p_Q\|_\infty \le C_3E_3^{-1/6}$, where $C_3 = 3\sqrt{2}\,e\,(9/2)^{1/6} < 15$.*
(b) *If $E_2 > 0$, then $\sup_{a\in\mathbb{R}}\mathbb{E}|Q-a|^{-1/2} \le C_2E_2^{-1/8}$, where $C_2 = 8e^{1/2}2^{-1/8} < 12.2$.*
(c) *If $E_1 = \|M\|_F^2 > 0$, then $|\phi_Q(t)| \le \big(\sqrt{2}\pi e/(\|M\|_F|t|)\big)^{1/2}$ for all $t \ne 0$.*

Proof. Write $\beta_\ell = ((\sqrt{2}\pi e)^\ell\omega(\ell)^{1/2})^{1/2}$, so that $|\phi_Q(t)| \le \min\{1, \beta_\ell E_\ell^{-1/4}|t|^{-\ell/2}\}$. (c) is the case $\ell = 1$. For (a), Lemma 2.2 with $p = 3/2$ and $a = \beta_3^{2/3}E_3^{-1/6}$ gives $\|p_Q\|_\infty \le 3a/\pi$, and $3\beta_3^{2/3}/\pi = 3\sqrt{2}e(9/2)^{1/6}$. For (b), Lemma 2.3 with $u = \beta_2E_2^{-1/4}$ gives $8\sqrt{u/(2\pi)}$, and $8(\beta_2/(2\pi))^{1/2} = 8e^{1/2}2^{-1/8}$ because $\beta_2 = 2^{3/4}\pi e$. □

5. One and two coordinates

Lemma 5.1. *Let Y have density $f \le 1$, let $a \neq 0$, and let $T = a(Y-m)^2 + v$. Then T has density*

$$p_T(z) = \frac{f(m+r)+f(m-r)}{2\sqrt{|a||z-v|}}\,\mathbf{1}\Big\{\frac{z-v}{a} > 0\Big\} \le \frac{1}{\sqrt{|a||z-v|}}, \qquad r = \sqrt{|z-v|/|a|}.$$

Moreover, for every interval U of length 2ρ and all real β, γ, the set $\{y : ay^2 + \beta y + \gamma \in U\}$ has Lebesgue measure at most $2\sqrt{2\rho/|a|}$.

Proof. The formula follows from the change of variables $y = m \pm r$ on the two half-lines on which $y \mapsto a(y-m)^2$ is monotone. For the second statement complete the square: the set in question is $\{y : (y-y_0)^2 \in U'\}$ for an interval U' of length $\delta = 2\rho/|a|$, and on each half-line $\{\pm(y-y_0) \ge 0\}$ it is an interval of length at most $\sqrt{u+\delta} - \sqrt{u} \le \sqrt{\delta}$ for some $u \ge 0$. □

Lemma 5.2. *Let $W \in \mathbb{R}^2$ have a density bounded by L, and let q be a quadratic polynomial on $\mathbb{R}^2$ whose quadratic part has a definite symmetric matrix B. Then $q(W)$ has a density bounded by $\pi L/\sqrt{\det B}$.*

Proof. Replacing q by $-q$ we may assume B is positive definite, and completing the square $q(w) = (w-w_0)^T B(w-w_0) + v$. Let $\mu_1, \mu_2 > 0$ be the eigenvalues of B with orthonormal eigenvectors e_1, e_2. The map $(\delta, \theta) \mapsto w_0 + \sqrt{\delta/\mu_1}\cos\theta\, e_1 + \sqrt{\delta/\mu_2}\sin\theta\, e_2$ is a diffeomorphism of $(0,\infty)\times[0,2\pi)$ onto $\mathbb{R}^2 \setminus \{w_0\}$ with $q = v + \delta$ and Jacobian $1/(2\sqrt{\det B})$. Hence $q(W)$ has density $(2\sqrt{\det B})^{-1}\int_0^{2\pi} p_W(w(\delta,\theta))\,\mathrm{d}\theta \le \pi L/\sqrt{\det B}$ at $v+\delta$, and density zero below v. This is a change of variables in a deterministic integral; no independence after rotation is used. □

Lemma 5.3. *Let W_1, W_2 be independent with densities $f_1, f_2 \le 1$, and let $q(w) = (w - w_0)^T B(w-w_0) + v$ with B real symmetric and indefinite, $D = |\det B| > 0$. Then $q(W)$ has a density satisfying, for almost every z,*

$$p_{q(W)}(z) \le D^{-1/2}\Big(6 + \log_+ \frac{16\sqrt{D}}{|z-v|}\Big). \tag{5.1}$$

Proof. Translating each coordinate separately preserves independence and the bounds on the densities, so we may assume $w_0 = 0$. Let $\mu_+ > 0 > \mu_-$ be the eigenvalues of B, with orthonormal eigenvectors e_+, e_-, and $\Lambda = \mu_+^{-1} + |\mu_-|^{-1}$. For $\delta > 0$ the level set $\{q = v+\delta\}$ is a hyperbola whose two branches are

$$w_\epsilon(\delta,\tau) = \epsilon\sqrt{\delta/\mu_+}\cosh\tau\, e_+ + \sqrt{\delta/|\mu_-|}\sinh\tau\, e_-, \qquad \epsilon \in \{1,-1\},\ \tau \in \mathbb{R}.$$

The map $(\epsilon,\delta,\tau) \mapsto w_\epsilon(\delta,\tau)$ is a bijection onto $\{w : q(w) > v\}$, and a direct computation gives Jacobian $1/(2\sqrt{D})$ in (δ,τ). Hence, for almost every $\delta > 0$,

$$p_{q(W)}(v+\delta) = \frac{1}{2\sqrt{D}}\sum_{\epsilon=\pm1}\int_{\mathbb{R}} f_1\big(w_{\epsilon,1}(\delta,\tau)\big)\, f_2\big(w_{\epsilon,2}(\delta,\tau)\big)\,\mathrm{d}\tau,$$

where $w_{\epsilon,1}, w_{\epsilon,2}$ are the coordinates in the original basis. Fix ϵ and write the branch as $\tau \mapsto ue^\tau + u'e^{-\tau}$ with $u = \frac12(\epsilon\sqrt{\delta/\mu_+}e_+ + \sqrt{\delta/|\mu_-|}e_-)$ and $u' = \frac12(\epsilon\sqrt{\delta/\mu_+}e_+ - \sqrt{\delta/|\mu_-|}e_-)$. By orthogonality $|u| = |u'| = \rho_0 := \frac12\sqrt{\delta\Lambda}$. Choose coordinates i and i' with $|u_i| \ge \rho_0/\sqrt{2}$ and $|u'_{i'}| \ge \rho_0/\sqrt{2}$.

For $\tau \ge T_1 = \frac34\log 2$ we have $|u'_i|e^{-\tau} \le \sqrt{2}|u_i|e^{-\tau} \le \frac12|u_i|e^{\tau}$, so the coordinate $y(\tau) = u_ie^\tau + u'_ie^{-\tau}$ keeps the sign of u_i, satisfies $\frac12|u_i|e^\tau \le |y(\tau)| \le \frac32|u_i|e^\tau$, and $|y'(\tau)| \ge \frac12|u_i|e^\tau \ge |y(\tau)|/3$. Let $T = \max\{T_1, \log(2/|u_i|)\}$ and $m = \frac12|u_i|e^T \ge 1$. On $[T,\infty)$ we bound the other factor by one, so $f_1f_2 \le f_i(y(\tau))$, and the change of variables $y = y(\tau)$ gives

$$\int_T^\infty f_1f_2\,\mathrm{d}\tau \le 3\int_{|y|\ge m,\ \text{one sign}} \frac{f_i(y)}{|y|}\,\mathrm{d}y \le 3\log\Big(1+\frac{1}{m}\Big) \le 3\log 2.$$

The middle inequality holds because, among functions with values in $[0,1]$ and integral at most one on the half-line $[m,\infty)$ (or $(-\infty,-m]$), the integral against the decreasing weight $1/|y|$ is largest for the indicator of the unit interval next to m. The same argument with u' and i' in place of u and i, and $\tau \to -\tau$, handles $(-\infty,-T']$ with $T' = \max\{T_1, \log(2/|u'_{i'}|)\}$; here $|u_{i'}|e^{\tau} \le \sqrt{2}|u'_{i'}|e^{\tau}$ plays the role of the earlier bound. Since $|u_i|$ and $|u'_{i'}|$ are at least $\sqrt{\delta\Lambda}/(2\sqrt{2})$, both cutoffs satisfy $T, T' \le \frac{3}{4}\log 2 + \frac{1}{2}\log_+(32/(\delta\Lambda))$, and on $[-T', T]$ the integrand is at most one. Each branch therefore contributes at most

$$6\log 2 + \tfrac{3}{2}\log 2 + \log_+ \frac{32}{\delta\Lambda} < 6 + \log_+ \frac{32}{\delta\Lambda}.$$

Finally $\Lambda \ge 2(\mu_+|\mu_-|)^{-1/2} = 2D^{-1/2}$ by the arithmetic–geometric mean inequality, so $32/(\delta\Lambda) \le 16\sqrt{D}/\delta$, and summing the two branches gives (5.1) for $z = v + \delta > v$. For $z < v$ apply the same argument to $-q$, whose matrix $-B$ has the same D. □

Lemma 5.4. *Let $S \neq 0$ be real symmetric, $L = \|S\|$, and $R = X^T S X + b^T X + c$, where X has independent coordinates with densities bounded by one. Then for all $a \in \mathbb{R}$ and $\rho > 0$,*

$$\mathbb{P}(|R - a| < \rho) \le \min\Bigl\{1, 19\sqrt{\rho/L}\Bigr\}. \tag{5.2}$$

In particular R has no atoms, and if $L \ge 1$ then

$$\sup_{a\in\mathbb{R}} \mathbb{E}\log_+ \frac{1}{|R-a|} \le L_0 := 2 + 2\log 19 < 8. \tag{5.3}$$

Proof. If $|S_{ii}| \ge L/2$ for some i, condition on the other coordinates: R becomes a quadratic polynomial in X_i with leading coefficient S_{ii}, and by Lemma 5.1 the conditional probability of $\{|R-a| < \rho\}$ is at most $2\sqrt{2\rho/|S_{ii}|} \le 4\sqrt{\rho/L}$.

Otherwise let O be the off-diagonal part of S. Then $\|O\| \ge \|S\| - \max_i |S_{ii}| \ge L/2$. For a uniformly random partition $[n] = I \sqcup J$, each ordered pair $i \neq j$ has $i \in I$, $j \in J$ with probability $1/4$, so $\mathbb{E}\|S_{I,J}\|_F^2 = \|O\|_F^2/4 \ge L^2/16$, and we fix a partition with $\|S_{I,J}\|_F \ge L/4$. Corollary 4.4(c) gives $|\phi_R(t)| \le (4\sqrt{2}\pi e/(L|t|))^{1/2}$. For $v > 0$ we have $\mathbf{1}\{|x| < \rho\} \le e^v e^{-vx^2/\rho^2}$, so by Lemma 2.1(c)

$$\begin{aligned}\mathbb{P}(|R-a| < \rho) &\le \frac{e^v \rho}{2\sqrt{\pi v}} \int_{\mathbb{R}} e^{-\rho^2 t^2/(4v)} |\phi_R(t)|\,\mathrm{d}t \\ &\le \frac{e^v \rho}{2\sqrt{\pi v}} \Bigl(\frac{4\sqrt{2}\pi e}{L}\Bigr)^{1/2} \Gamma\bigl(\tfrac{1}{4}\bigr) \Bigl(\frac{4v}{\rho^2}\Bigr)^{1/4},\end{aligned}$$

using $\int_{\mathbb{R}} e^{-wt^2}|t|^{-1/2}\,\mathrm{d}t = \Gamma(1/4) w^{-1/4}$. At $v = 1/4$ this equals

$$2e^{1/4}(\sqrt{2}\pi e)^{1/2}\,\Gamma(1/4)\,\pi^{-1/2}\sqrt{\rho/L} < 19\sqrt{\rho/L}.$$

This proves (5.2); letting $\rho \to 0$ excludes atoms. If $L \ge 1$, then by the layer-cake formula

$$\mathbb{E}\log_+ \frac{1}{|R-a|} = \int_0^\infty \mathbb{P}(|R-a| < e^{-u})\,\mathrm{d}u \le \int_0^\infty \min\{1, 19e^{-u/2}\}\,\mathrm{d}u = 2\log 19 + 2. \quad \square$$

Lemma 5.5. *For $d \neq 0$,*

$$\mathcal{J}(d) := \int_{-1}^{1} |x|^{-1/2}|x-d|^{-1/2}\,\mathrm{d}x \le \pi + 4\log 2 + 2\log_+ \frac{1}{|d|}, \tag{5.4}$$

and for every $a \in \mathbb{R}$,

$$\int_{-1}^{1} |x|^{-1/2} \log_+ \frac{1}{|x-a|}\,\mathrm{d}x \le 8. \tag{5.5}$$

Proof. Since $\mathcal{J}(-d) = \mathcal{J}(d)$ we may take $d > 0$. Let first $d \le 1$. The functions $2\log(\sqrt{y} + \sqrt{y+d})$, $2\arcsin\sqrt{x/d}$ and $2\log(\sqrt{x} + \sqrt{x-d})$ are antiderivatives of $y^{-1/2}(y+d)^{-1/2}$ on

$(0,1)$, of $x^{-1/2}(d-x)^{-1/2}$ on $(0,d)$ and of $x^{-1/2}(x-d)^{-1/2}$ on $(d,1)$. Splitting the integral at 0 and d, and putting $x=-y$ on $[-1,0]$, we get

$$\begin{aligned}\mathcal{J}(d) &= 2\log\frac{1+\sqrt{1+d}}{\sqrt{d}} + \pi + 2\log\frac{1+\sqrt{1-d}}{\sqrt{d}} \\ &= \pi + 2\log\bigl((1+\sqrt{1+d})(1+\sqrt{1-d})\bigr) + 2\log\frac{1}{d}.\end{aligned}$$

The product equals $1+\sqrt{1+d}+\sqrt{1-d}+\sqrt{1-d^2}\le 4$, because $\sqrt{1+d}+\sqrt{1-d}\le 2$ by concavity of the square root. If $d>1$, then $|x-d| = d-x \ge 1-x = |x-1|$ for $x\in[-1,1]$, so $\mathcal{J}(d)\le \mathcal{J}(1) = \pi + 2\log(1+\sqrt{2}) < \pi + 4\log 2$. For (5.5), write both nonnegative factors as integrals of the indicators of their superlevel sets, which are intervals centred at 0 and at a respectively. The length of the intersection of two intervals is at most the length of the shorter one, with equality when they have the same centre, so by Tonelli's theorem the integral is largest at $a=0$, where it equals $2\int_0^1 x^{-1/2}\log(1/x)\,\mathrm{d}x = 8$. □

Lemma 5.6. *Let W_1, W_2 be independent with densities bounded by one, and let q be a quadratic polynomial on $\mathbb{R}^2$ whose quadratic part has a symmetric matrix B with $D = |\det B| > 0$. Then*

$$\sup_{a\in\mathbb{R}} \mathbb{E}|q(W)-a|^{-1/2} \le 16 D^{-1/4}.$$

Proof. If a random variable Z has a density bounded by M, then for every $a\in\mathbb{R}$

$$\mathbb{E}|Z-a|^{-1/2} = \int_0^\infty \mathbb{P}(|Z-a|<s^{-2})\,\mathrm{d}s \le \int_0^\infty \min\{1, 2Ms^{-2}\}\,\mathrm{d}s = 2\sqrt{2M}. \tag{5.6}$$

If B is definite, Lemma 5.2 gives $M=\pi D^{-1/2}$ and the bound $2\sqrt{2\pi}\,D^{-1/4} < 6D^{-1/4}$. Let B be indefinite, write $q(w) = (w-w_0)^T B(w-w_0)+v$, and let $Y = (q(W)-v)/\sqrt{D}$. By Lemma 5.3, Y has a density bounded by $h(y) = 6+\log_+(16/|y|)$. Fix $b\in\mathbb{R}$ and let $\delta = 1/64$. The contribution of the event $\{|Y-b|>\delta\}$ to $\mathbb{E}|Y-b|^{-1/2}$ is at most $\delta^{-1/2}=8$, and the rest is at most

$$\int_{|y-b|\le\delta} h(y)|y-b|^{-1/2}\,\mathrm{d}y \le \int_{-\delta}^{\delta} h(y)|y|^{-1/2}\,\mathrm{d}y = 4\sqrt{\delta}\Bigl(8+\log\frac{16}{\delta}\Bigr) = 4+5\log 2.$$

The inequality is the rearrangement argument from the proof of Lemma 5.5: the superlevel sets of h and of $|y-b|^{-1/2}\mathbf{1}\{|y-b|\le\delta\}$ are intervals centred at 0 and at b, or the whole line. Hence $\mathbb{E}|Y-b|^{-1/2} \le 12+5\log 2 < 16$ for every b, and $\mathbb{E}|q(W)-a|^{-1/2} = D^{-1/4}\,\mathbb{E}|Y-(a-v)/\sqrt{D}|^{-1/2}$. □

6. Random signs

Throughout this section A is a real symmetric $n\times n$ matrix and

$$\eta = \frac{1}{12}, \qquad \zeta = \frac{1}{256}, \qquad a_0 = \frac{5}{48}. \tag{6.1}$$

Definition 6.1. A pivot of A is either an index i with $|a_{ii}|\ge\eta$ or a pair $i\ne j$ with $|a_{ii}a_{jj}-a_{ij}^2|\ge\zeta$.

If A has no pivot, then for $i\ne j$ we have $a_{ij}^2 = a_{ii}a_{jj} - (a_{ii}a_{jj}-a_{ij}^2) < \eta^2+\zeta = \frac{25}{2304} = a_0^2$, so

$$|a_{ii}|<\eta, \qquad |a_{ij}|<a_0 \quad (i\ne j). \tag{6.2}$$

Lemma 6.2. *Let $[n] = I\sqcup J$ be a partition, and let C be the symmetric matrix with $C_{ij}=a_{ij}$ when i and j lie in different parts and $C_{ij}=0$ otherwise.*

(a) *If $\varepsilon_i = 1$ for $i\in I$, $\varepsilon_i=-1$ for $i\in J$, and $D=\operatorname{diag}(\varepsilon)$, then $A-DAD = 2C$.*
(b) *For every $m\ge 1$, $s_{2m-1}(C) = s_{2m}(C) = s_m(A_{I,J})$.*

Proof. (a) The (i,j) entry of DAD is $\varepsilon_i\varepsilon_j a_{ij}$, which equals a_{ij} when i and j lie in the same part and $-a_{ij}$ otherwise. (b) Order the coordinates so that I comes first; then $C = \begin{pmatrix} 0 & M \\ M^T & 0 \end{pmatrix}$ with $M = A_{I,J}$. Let $M = \sum_k \sigma_k u_k v_k^T$ be a singular value decomposition, with $\sigma_k > 0$ and orthonormal families (u_k) and (v_k). The vectors $(u_k, \pm v_k)/\sqrt{2}$ are orthonormal eigenvectors of C with eigenvalues $\pm\sigma_k$. Their orthogonal complement consists of the vectors (x,y) with x orthogonal to the range of M and y orthogonal to the range of M^T, that is, with $M^T x = 0$ and $My = 0$, and on it $C(x,y) = (My, M^T x) = 0$. Hence the singular values of C are the σ_k, each counted twice, followed by zeros. □

Lemma 6.3. *Let O be a real symmetric matrix with zero diagonal and $|o_{ij}| \le a$ for all $i \ne j$, let $\phi_1, \dots, \phi_n \in \mathbb{R}^k$, and for $\varepsilon \in \{-1,1\}^n$ put*

$$Q_\varepsilon = \sum_{i \ne j} \varepsilon_i \varepsilon_j \, o_{ij} \, \phi_i \phi_j^T \in \mathbb{R}^{k \times k}.$$

If ε is uniformly distributed on $\{-1,1\}^n$, then

$$\mathbb{E}\|Q_\varepsilon\|_F^2 = \sum_{i \ne j} o_{ij}^2 \big(|\phi_i|^2|\phi_j|^2 + (\phi_i \cdot \phi_j)^2\big) \le 2a^2 \Big(\sum_{i=1}^n |\phi_i|^2\Big)^2.$$

Proof. With $\langle X, Y\rangle = \operatorname{tr}(X^T Y)$ we have $\langle \phi_i\phi_j^T, \phi_p\phi_q^T\rangle = (\phi_i \cdot \phi_p)(\phi_j \cdot \phi_q)$, so

$$\|Q_\varepsilon\|_F^2 = \sum_{i \ne j} \sum_{p \ne q} \varepsilon_i\varepsilon_j\varepsilon_p\varepsilon_q \, o_{ij} o_{pq} \, (\phi_i \cdot \phi_p)(\phi_j \cdot \phi_q).$$

The ε_i are independent with mean zero and $\varepsilon_i^2 = 1$. Let $i \ne j$ and $p \ne q$. If $\{p,q\} = \{i,j\}$, then $\varepsilon_i\varepsilon_j\varepsilon_p\varepsilon_q = 1$. Otherwise one of p, q, say p, lies outside $\{i,j\}$; since also $p \ne q$, the index p occurs exactly once in the product, whose expectation is therefore zero. The pairs $(p,q) = (i,j)$ and $(p,q) = (j,i)$ contribute $o_{ij}^2|\phi_i|^2|\phi_j|^2$ and $o_{ij}o_{ji}(\phi_i \cdot \phi_j)^2 = o_{ij}^2(\phi_i \cdot \phi_j)^2$, which gives the identity. The inequality follows from $(\phi_i \cdot \phi_j)^2 \le |\phi_i|^2|\phi_j|^2$, $o_{ij}^2 \le a^2$ and $\sum_{i \ne j}|\phi_i|^2|\phi_j|^2 \le (\sum_i |\phi_i|^2)^2$. □

Proposition 6.4. *Let $m \in \{1,2,3\}$ and*

$$\gamma_m = \frac{44 - 5\sqrt{2}\,(2m-1)}{96}, \qquad \text{so that} \quad \gamma_1 > 0.38, \quad \gamma_2 > 0.237, \quad \gamma_3 > 0.09.$$

If A has no pivot and $s_{2m-1}(A) \ge 1$, then there is a partition $[n] = I \sqcup J$ with $s_m(A_{I,J}) > \gamma_m$, and consequently $E_m(A_{I,J}) > \gamma_m^{2m}$.

Proof. Put $k = 2m-1$. Since $s_k(A) \ge 1$, A has orthonormal eigenvectors $\varphi_1, \dots, \varphi_k$ with eigenvalues $\lambda_1, \dots, \lambda_k$ of modulus at least one. Let $\Phi \in \mathbb{R}^{n \times k}$ have columns $\varphi_1, \dots, \varphi_k$, so that $\Phi^T\Phi = I_k$ and $\Phi^T A \Phi = \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_k)$, and let $\phi_i \in \mathbb{R}^k$ be the i-th row of Φ; then $\sum_i |\phi_i|^2 = \operatorname{tr}(\Phi^T\Phi) = k$. Write $A = \Delta + O$ with Δ diagonal and O vanishing on the diagonal. For $\varepsilon \in \{-1,1\}^n$ and $D = \operatorname{diag}(\varepsilon)$ we have $D\Delta D = \Delta$ and $\Phi^T D O D \Phi = Q_\varepsilon$, with Q_ε as in Lemma 6.3. By (6.2), the largest modulus a of an off-diagonal entry of A satisfies $a < a_0$, so by Lemma 6.3 some ε has $\|Q_\varepsilon\|_F^2 \le 2a^2k^2$ and hence $\|Q_\varepsilon\|_F < \sqrt{2}\,a_0 k$. Let $I = \{i : \varepsilon_i = 1\}$, $J = [n] \setminus I$, and let C be as in Lemma 6.2. By Lemma 6.2(a),

$$2\,\Phi^T C \Phi = \Phi^T (A - DAD)\Phi = \Lambda - \Phi^T \Delta \Phi - Q_\varepsilon.$$

Here $\|\Phi^T\Delta\Phi\| \le \max_i |a_{ii}| < \eta$ and $\|Q_\varepsilon\| \le \|Q_\varepsilon\|_F < \sqrt{2}\,a_0 k$, while $s_k(\Lambda) = \min_j |\lambda_j| \ge 1$. By Lemma 2.5,

$$s_k(2\,\Phi^T C \Phi) > 1 - \eta - \sqrt{2}\,a_0 k = \frac{44 - 5\sqrt{2}\,k}{48} = 2\gamma_m,$$

and, again by Lemma 2.5, $s_k(C) \ge s_k(\Phi^T C \Phi) > \gamma_m$. By Lemma 6.2(b), $s_m(A_{I,J}) = s_{2m-1}(C) = s_k(C) > \gamma_m$. Finally, by Lemma 2.7(a), $E_m(A_{I,J}) = e_m(MM^T)$ with $M = A_{I,J}$

is a sum of products of m of the eigenvalues $s_j(M)^2$ of MM^T, all nonnegative, and one of these products is $\prod_{j\le m} s_j(M)^2 > \gamma_m^{2m}$.

The numerical bounds follow from $\sqrt{2} < 1.4143$: $\gamma_1 > (44 - 7.0715)/96 > 0.38$, $\gamma_2 > (44 - 21.2145)/96 > 0.237$ and $\gamma_3 > (44 - 35.3575)/96 > 0.09$. □

7. Proofs of Theorems 1.1 and 1.2

In this section X has independent coordinates with densities bounded by one, and η, ζ are as in (6.1).

Proposition 7.1. *Suppose that A has no pivot, and let $Q = X^T AX + b^T X + c$.*

(a) *If $s_5(A) \ge 1$, then Q has a continuous density bounded by $C_3/\gamma_3 < 167$.*

(b) *If $s_3(A) \ge 1$, then $\sup_a \mathbb{E}|Q - a|^{-1/2} \le C_2\gamma_2^{-1/2} < 26$.*

Proof. (a) By Proposition 6.4 with $m = 3$ some partition has $E_3(A_{I,J}) > \gamma_3^6$, so $E_3^{-1/6} < \gamma_3^{-1} < 100/9$. Apply Corollary 4.4(a): the density is at most $15 \cdot 100/9 < 167$. (b) By Proposition 6.4 with $m = 2$ some partition has $E_2(A_{I,J}) > \gamma_2^4$, so $E_2^{-1/8} < \gamma_2^{-1/2} < 0.237^{-1/2} < 2.055$. Apply Corollary 4.4(b): $12.2 \cdot 2.055 < 26$. □

Proposition 7.2. *If $s_3(A) \ge 1$, then $\sup_{a\in\mathbb{R}} \mathbb{E}|Q - a|^{-1/2} \le 77$ for $Q = X^T AX + b^T X + c$ and all b, c.*

Proof. If A has no pivot, Proposition 7.1(b) gives the bound 26.

A pair. Suppose that $|a_{ii}a_{jj} - a_{ij}^2| \ge \zeta$ for some $i \ne j$. Conditionally on the coordinates other than X_i and X_j, Q is a quadratic polynomial in (X_i, X_j) whose quadratic part has the matrix $A_{\{i,j\},\{i,j\}}$, so by Lemma 5.6 its conditional negative half moment about any a is at most $16\zeta^{-1/4} = 64$. Averaging over the other coordinates gives the same bound for Q.

A diagonal pivot. Suppose that $|a_{ii}| \ge \eta$. Completing the square in X_i,

$$Q = a_{ii}\big(X_i - \mu(X_{-i})\big)^2 + R(X_{-i}), \tag{7.1}$$

where μ is affine and R is a quadratic polynomial in X_{-i} whose matrix is the Schur complement $A_{-i,-i} - A_{-i,i}A_{i,-i}/a_{ii}$. By Lemma 5.1, conditionally on X_{-i} the density of Q is at most $|a_{ii}|^{-1/2}|z - R(X_{-i})|^{-1/2}$. By Lemma 2.4, for every a,

$$\mathbb{E}\big[|Q - a|^{-1/2}\mathbf{1}\{|Q - a| \le 1\}\big] \le |a_{ii}|^{-1/2}\,\mathbb{E}\,\mathcal{J}\big(R(X_{-i}) - a\big).$$

By Lemma 2.6 the Schur complement has at least two eigenvalues of modulus at least one, so its norm is at least one. By Lemma 5.4, R has no atoms and $\mathbb{E}\log_+(1/|R - a|) \le L_0$, and (5.4) bounds the right side by $\eta^{-1/2}(\pi + 4\log 2 + 2L_0)$. Since $\pi + 4\log 2 + 2L_0 = \pi + 4 + 4\log 2 + 4\log 19 < 21.92$, by $\pi < 3.1416$, $\log 2 < 0.6932$ and $\log 19 < 3$, and $\eta^{-1/2} = \sqrt{12} < 3.465$, this is less than 76. As $|Q - a|^{-1/2} \le 1$ when $|Q - a| > 1$, it follows that $\mathbb{E}|Q - a|^{-1/2} < 77$. □

Proof of Theorem 1.1. First let $K = s = 1$, so that A has at least five eigenvalues of modulus at least one.

If A has no pivot, Proposition 7.1(a) bounds the density by 167.

If $|a_{ii}| \ge \eta$ for some i, write Q as in (7.1). By Lemma 2.6, the matrix of R has at least four eigenvalues of modulus at least one, and in particular $s_3 \ge 1$. Lemmas 5.1 and 2.4 and Proposition 7.2, applied to R as a quadratic polynomial in the independent coordinates X_{-i}, give for almost every z

$$p_Q(z) \le |a_{ii}|^{-1/2}\,\mathbb{E}|z - R(X_{-i})|^{-1/2} \le \eta^{-1/2} \cdot 77 = \sqrt{12} \cdot 77 < 267.$$

Suppose finally that all diagonal entries have modulus less than η and that some pair $\{i, j\}$ has $|a_{ii}a_{jj} - a_{ij}^2| \ge \zeta$. Conditionally on the other coordinates, Q is a quadratic polynomial in (X_i, X_j) with quadratic part $A_{\{i,j\},\{i,j\}}$. If this matrix is definite, Lemmas 5.2 and 2.4 bound the density of Q by $\pi\zeta^{-1/2} = 16\pi < 51$. If it is indefinite, with $D = a_{ij}^2 - a_{ii}a_{jj} \ge \zeta$, the

conditional critical value v is a quadratic polynomial in the other coordinates whose matrix is the Schur complement of the pair; by Lemma 2.6 it has at least three eigenvalues of modulus at least one, so Lemma 5.4 gives $\mathbb{E}\log_+(1/|z-v|) \le L_0$ for every z. By (5.1), the inequality $\log_+(xy) \le \log_+ x + \log_+ y$ and Lemma 2.4, for almost every z

$$p_Q(z) \le D^{-1/2}\bigl(6 + \log_+(16\sqrt{D}) + L_0\bigr) \le 16\bigl(8 + 4\log 2 + 2\log 19\bigr) + \frac{1}{e} < 269,$$

where we used $\log_+(16\sqrt{D}) \le \log 16 + \frac{1}{2}\log_+ D$, $D \ge \zeta$ and $\sup_{D>0} D^{-1/2}\log_+ D = 2/e$, together with $\log 2 < 0.6932$, $\log 19 < 3$ and $e > 2.7$. In every case $\|p_Q\|_\infty < 269$.

For general K and s, the vector $Y = KX$ has independent coordinates with densities bounded by one, and

$$Q = \frac{s}{K^2}\,\widetilde{Q}, \qquad \widetilde{Q} = Y^T\frac{A}{s}\,Y + \frac{K}{s}\,b^TY + \frac{K^2}{s}\,c.$$

Since $s_5(A/s) \ge 1$, $\widetilde{Q}$ has a density less than 269, and $p_Q(z) = (K^2/s)\,p_{\widetilde{Q}}(K^2z/s) < 269\,K^2/s$. The interval form follows by integrating the density, and conversely a bound $\mathbb{P}(Q \in J) \le C|J|$ for all intervals extends to open sets and then, by outer regularity, to all Borel sets. □

Proof of Theorem 1.2. With Y and $\widetilde{Q}$ as in the previous proof and $s_3(A/s) \ge 1$, Proposition 7.2 gives, for every a,

$$\mathbb{E}|Q-a|^{-1/2} = \frac{K}{\sqrt{s}}\,\mathbb{E}\Bigl|\widetilde{Q} - \frac{K^2a}{s}\Bigr|^{-1/2} \le \frac{77\,K}{\sqrt{s}}.$$ □

8. Remarks on the threshold

Remark 8.1. Three is the least number of large eigenvalues for which a bound of the form (1.1) can hold. With X_1, X_2 uniform on $[0,1]$ the forms X_1^2 and $X_1^2 - X_2^2$ have unbounded densities, and $A = \operatorname{diag}(\lambda, 0, \dots)$ or $\operatorname{diag}(\lambda, -\lambda, 0, \dots)$ may be scaled to make λ as large as we like without improving the density near the singular value. Whether three eigenvalues of modulus at least s always suffice, with a bound CK^2/s for an absolute C, is open. For diagonal matrices the answer is yes with $C = \pi$, by the argument in the introduction.

Remark 8.2. The number five is used only in the case without pivots, through Proposition 6.4 with $m = 3$; the diagonal pivot uses four large eigenvalues of A (three survive in the Schur complement, as Proposition 7.2 requires), and the pair cases use three. Four eigenvalues without pivots do not give a third large singular value in any off-diagonal block. Let $N > 256$, let $w \in \mathbb{R}^N$ have all entries equal to $N^{-1/2}$, and let A be the direct sum of two copies of the $(N+1)\times(N+1)$ matrix $\begin{pmatrix} 0 & w^T \\ w & 0 \end{pmatrix}$. Its nonzero eigenvalues are $1, 1, -1, -1$, its diagonal vanishes and every principal 2×2 minor has modulus at most $1/N < \zeta$, so A has no pivot. For every partition $[n] = I \sqcup J$ the part of each copy in $A_{I,J}$ lies in a single row or a single column, namely that of the centre of the copy, so $A_{I,J}$ has rank at most two and $E_3(A_{I,J}) = 0$. For this matrix the density of Q is nevertheless bounded, whatever b and c are. Write U_1, U_2 for the two centres and V_k for the vector of the other coordinates of the k-th copy, so that $Q = Z_1 + Z_2 + c$ with independent summands $Z_k = U_k(2w\cdot V_k + \alpha_k) + \beta_k \cdot V_k$, where $\alpha_k \in \mathbb{R}$ and $\beta_k \in \mathbb{R}^N$ collect the entries of b. Given $U_k = u$, the variable $Z_k - \alpha_k u$ is the linear form $v(u)\cdot V_k$ with $v(u) = 2uw + \beta_k$, which by Lemma 3.2 has a density $g_u \le \sqrt{e}/|v(u)|$; hence $\|\widehat{g_u}\|_2^2 = 2\pi\|g_u\|_2^2 \le 2\pi\sqrt{e}/|v(u)|$. Write $\beta_k = rw + \beta_k^\perp$ with $\beta_k^\perp$ orthogonal to w; since $|w| = 1$, $|v(u)| \ge |2u + r|$. With $f \le 1$ the density of U_k, $\phi_{Z_k}(t) = \int f(u)e^{it\alpha_k u}\widehat{g_u}(t)\,\mathrm{d}u$, and Minkowski's integral inequality gives $\|\phi_{Z_k}\|_2 \le \sqrt{2\pi}\,e^{1/4}\int f(u)|2u+r|^{-1/2}\,\mathrm{d}u \le 2\sqrt{2\pi}\,e^{1/4}$, the last integral being largest when f is the indicator of the unit interval centred at $-r/2$. By the Cauchy–Schwarz inequality $\|\phi_Q\|_1 \le 8\pi\sqrt{e}$, and Lemma 2.1(b) bounds the density of Q by $4\sqrt{e}$. We do not know whether four large eigenvalues always suffice.

Remark 8.3. Order three is the smallest order at which Theorem 4.1 gives an integrable envelope, and the second crossing volume alone does not bound the density. Let X_1, X_2, Y_1, Y_2 be independent and uniform on $[-\frac{1}{2}, \frac{1}{2}]$, and $Q = (X_1 + Y_1)^2 - (X_2 + Y_2)^2$. For the partition $I = \{X_1, X_2\}$, $J = \{Y_1, Y_2\}$ the cross block is $\operatorname{diag}(1, -1)$, so $E_2 = 1$. But $U = X_1 + Y_1$ and $V = X_2 + Y_2$ are independent with the triangular density on $[-1, 1]$, and U^2 and V^2 have densities $(1 - \sqrt{y})/\sqrt{y}$ on $(0, 1)$, so the density of $Q = U^2 - V^2$ at 0 is $\int_0^1 (1 - \sqrt{y})^2 y^{-1} \, \mathrm{d}y = \infty$; near 0 it grows like $\log(1/|z|)$. By Theorem 1.2, or by Theorem 4.1 at order two, the negative half moment of this Q is nevertheless bounded.

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel